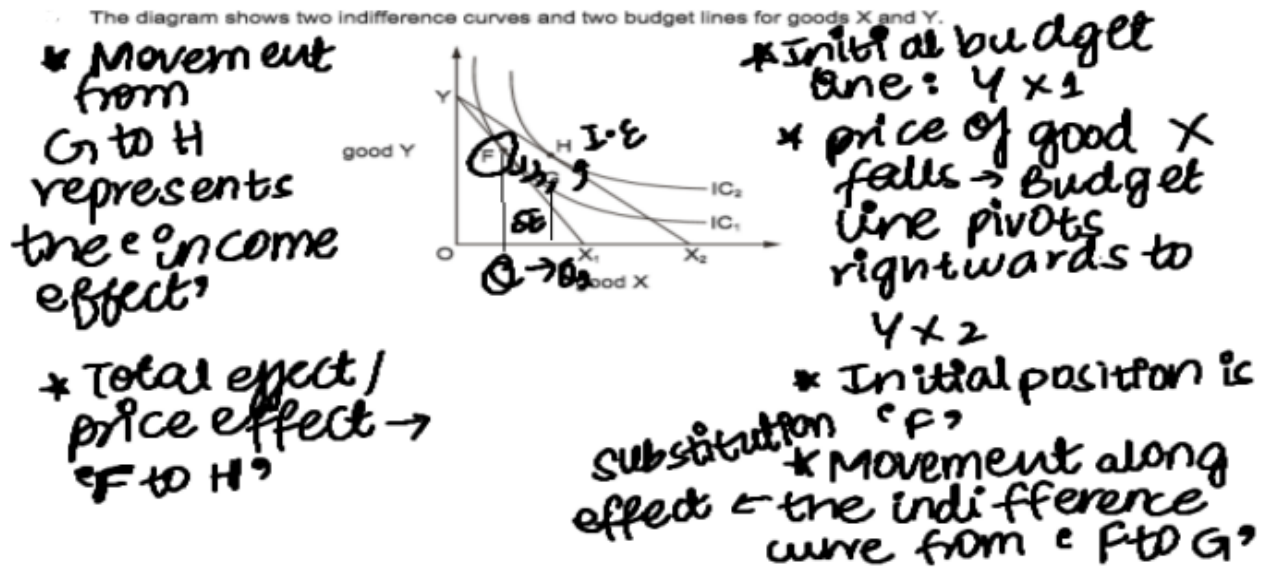


Free market equilibrium: $e \rightarrow MP_C = MP_B$

4-A:



A – The income effect for a price fall for X is shown from a movement from G to H. This is why, this is correct.

B- The price effect for a price change for X is the total effect from F to H, since a price effect includes both: income effect and the substitution effect.

C – The substitution effect for a price fall for X should be a movement along the indifference curve from "F to G"

D – The substitution effect for a price rise for X should bring a contraction in quantity demanded of X.

5-A:

A – In an "eat all you can banquet", if a person consumes one food consecutively from all the courses, then that particular food provides less and less satisfaction than before, as the quantity consumed increases.

B – This is the limitation of the utility theory, that in some cases, the amount of satisfaction gained from additional units of consumption of a product might increase as the number consumed increases. This may be in case of coin collectors. A collector wishing to have the largest coin collection will obtain the greatest satisfaction from the "last coin purchased"

C -The second of the two rolls of wallpaper needed to decorate a room. This option is incorrect because the consumption of two rolls of wallpaper does not necessarily reflect the concept of diminishing marginal utility. Each roll of wallpaper is likely to provide similar utility in terms of covering the required area and achieving the desired outcome. The diminishing marginal utility concept focuses on the diminishing satisfaction gained from consuming additional units of a particular good or service.

D: This option is incorrect because the law of diminishing marginal utility does not directly apply to the driving lessons needed to pass a driving test. Each driving lesson may contribute to the learner's progress and improvement in driving skills, and there is no inherent diminishing satisfaction associated with taking additional lessons.

6-A:

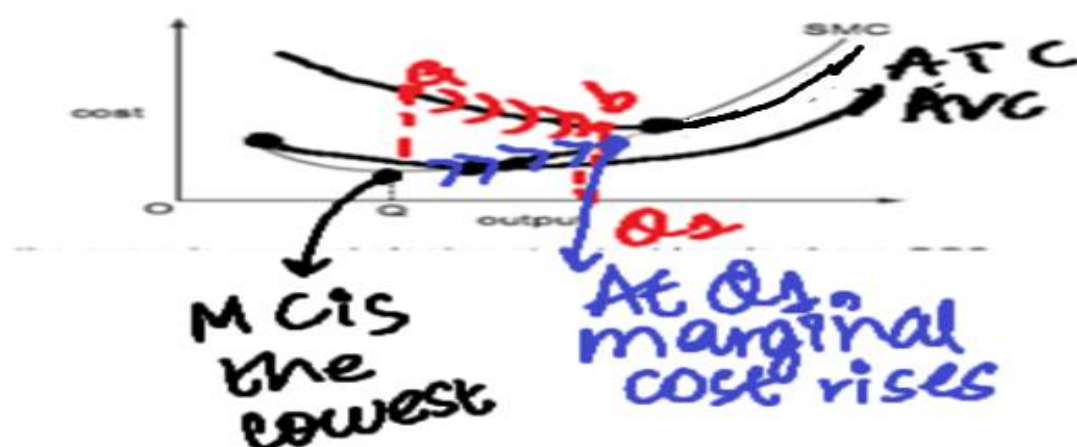
A – If firms achieve minimum efficient to scale point at a smaller level of output, and beyond that they experience “dis-economies of scale”, that is cost disadvantages in form of higher long run average costs, they are not likely to expand, and may operate at a smaller level of output. This is why, this becomes correct.

B- If government policy encourages horizontal integration, firms are likely to expand and may gain higher market share as they become large

C- If integration (external growth) is impossible, firms still have the option to grow internally, and they may still become large.

D – If there is an international market for the product, firms are likely to expand to grow internationally, and there is a likelihood of large firms existing.

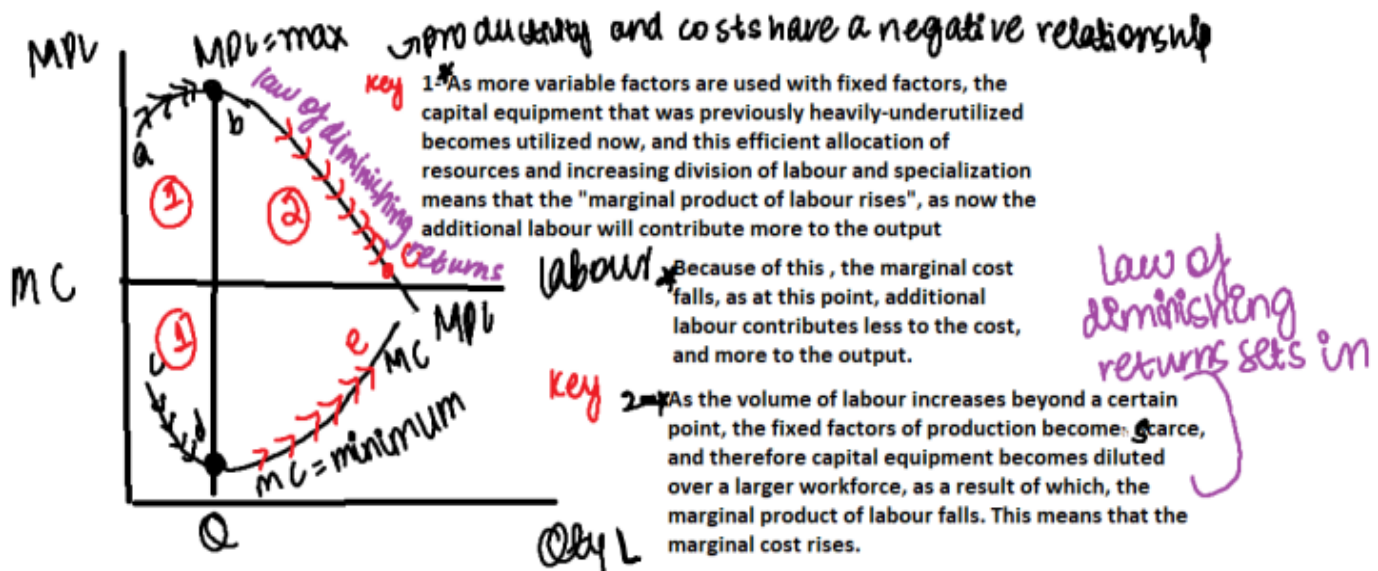
7-D:



A- Diseconomies of scale is associated with a rise in Average total cost. At output levels above OQ , for example, OQ_1 , the average total cost is seen to fall (this can be seen in the first diagram above)

B- Inelasticity of supply is not linked to this.

C- This is incorrect as fixed costs do not change in a given time period.



D- It can be seen above (in the first figure) that an increase in output from Q to Q_1 results in a rise in marginal costs. This is because, the law of variable proportions or the law of diminishing returns sets in, since a rise in marginal costs is a result of a fall in marginal product of labor. (This can be seen in the second figure-the diagram of "mirror picture", and is explained in key 2). This is why, key D becomes correct.

8-B:

A- The firm is not operating with diseconomies of scale, it is experiencing "economies of scale" as the average total cost is falling.

B- Yes, it can be seen above that the firm's average variable cost is constant and of \$3 at each level of output. This explains that key B is correct.

C- Fixed cost is the cost that remains unchanged over a range of activity. At zero level of output, fixed cost = total cost, and that is "9"

D- We have no information regarding revenues.

TC	9	12	15	18	21	24
output	0	1	2	3	4	5
average total cost \$		12	7.5	6	5.25	4.8
marginal cost \$		3	3	3	3	3

$$ATC = \frac{TC}{Q}$$

- MC (change in TC)
- $TC_1 - TC_0 = MC_1$

VC
note!

$$VC = \frac{VC}{Q}$$

3 6 9 12 15

$$ATC \times Q = TC$$

3 3 3 3 3

3/1 6/2 9/3 12/4 15/5

$$12 - x = 3$$

$$x = 9$$

total cost at "0" level of output $\rightarrow 9 \rightarrow$ fixed cost

$$VC = TC - FC$$

9-D:

The principal-agent problem refers to a situation where there is a conflict of interest between the owner (principal) and the manager (agent) of a firm. **The principal-agent problem arises because the owner delegates decision-making authority to the manager, but there may be a divergence in their goals and objectives.**

Option A: The disincentive effect for entrepreneurs of high government tax rates. This option is not an example of the principal-agent problem. It refers to the disincentive effect of high tax rates on entrepreneurs, which is related to the impact of taxation on entrepreneurship rather than a conflict between owners and managers.

Option B: The existence of a trade union to put forward workers' views to managers. This option is not an example of the principal-agent problem. It refers to the presence of a trade union representing workers' interests and advocating for their views, which is more related to the dynamics between workers and management **rather than the conflict between owners and managers.**

Option C: The lack of consumer knowledge of the quality of firms' products. This option is not an example of the principal-agent problem. It relates to the information asymmetry between consumers and firms regarding the quality of products, which is associated with issue of imperfect information and consumer protection rather **than conflicts between owners and managers.**

Option D: The separation of the owners of a firm from the firm's manager. This option is correct because it exemplifies the principal-agent problem. When the owners (shareholders) of a firm are separate from the firm's manager, they delegate decision-making authority to the manager to run the company on their behalf. However, the manager may have different goals than the owners, leading to a potential conflict of interest. The manager may prioritize **sales revenue maximization because their salaries are based on revenues** or pursue objectives that do not align with shareholders' objective of **profit maximization**

10-A:

A – Higher long-run profits can never be achieved in monopolistic competition, as due to freedom of entry and exit, firms only make “normal profits in the long run”. Therefore, option A, becomes correct.

B – In the short-run, when increased advertising improves the brand image of the product, and makes it appear more differentiated, demand will increase, and so, higher selling price can be charged.

C – Advertising helps build strong brand identities for products. Brands act as a signal to consumers about the quality, reputation, and reliability of a product. A well-established brand can create loyalty and preference among consumers, leading them to perceive the product as **different** and better than its competitors.

Note:- By associating positive emotions or experiences with their products, firms can create a unique and differentiated image in consumers' minds.

D – By building “brand-loyalty” advertising can change elasticity from “elastic” to “inelastic”

11-A:

Note: In a contestable market structure, a monopoly being a monopoly cannot act as a monopoly due to potential threat of competition arising from freedom of entry and exit.

A - A contestable market: This option is correct because it captures the essence of a market structure where a small number of existing firms face competition from potential entrants. In a contestable market, entry barriers are low, allowing new firms to enter and compete with existing firms. This potential competition leads to the firms behave as if they are operating in actual competition, even if there are only a few players in the market.

B: A monopoly refers to a market structure where there is a single dominant firm with a market share of about or over 25%, and it has significant market power and faces very limited competition. In a monopoly, entry barriers are high, preventing potential entrants from entering and competing in the market.

C: Monopolistic competition. This option is incorrect. Monopolistic competition describes a market structure where there are many firms (mainly 20-25 small firms) that produce differentiated products, and each firm has some degree of market power. In monopolistic competition, firms compete based on product differentiation, but the ease of entry for potential competitors may still be limited.

D: If the firms in an oligopoly market structure manufacture homogenous products, then it is known as a pure or a perfect oligopoly. Even though it is rare to find oligopoly firms with homogeneous products, industries such as steel, cement, aluminum come under pure oligopoly.

12-C:

At output of "OX", total revenue is shown to be maximized that is when Marginal revenue is equal to zero. This means that the objective of the firm is "sales-revenue maximization", which is why, key C is correct.

That being said, other objectives such as growth-rate maximization (key A), long-run profit-maximization (key B), short-run profit maximization (key D) are all associated with profit-maximization in a way, and this means that the firm would operate at a profit maximizing level of output (where Marginal cost = Marginal Revenue).

The firm would not operate where Total Revenue = Total cost, that is when "normal profits are made", if it wants to maximize its profits. This is why, these options become incorrect.

13-D:

A – This is incorrect because due to low barriers to entry, the firms in monopolistic competition are not able to prevent new firms from entering the industry.

B- Although, it is correct to say that the firms produce where Average costs are falling, and so experience "excess capacity", this statement becomes incorrect because this does not influence price.

C – This is incorrect because in an imperfect competition, Average revenue is always greater than marginal revenue.

D – Product differentiation affects the price elasticity of demand for each product. Products that are perceived as more unique and indispensable to consumers are likely to have less elastic demand, allowing firms to charge higher prices without losing significant market share. On the other hand, products with close substitutes and weaker differentiation may face more elastic demand, limiting the firm's ability to raise prices without losing customer.

Instead of engaging in direct price competition, firms in monopolistic competition often focus on non-price competition. They invest in advertising, product development, and customer service to enhance the perceived value of their products and attract consumers based on factors other than price. This approach allows firms experience a rise in demand and to maintain higher prices.

14-C: C) Large-scale poverty is not a justification for state intervention on grounds of market failure. Market failure occurs when the free market fails to allocate resources efficiently from the "society's point of view", leading to suboptimal outcomes. However, the existence of poverty is not an example of "market failure". Instead, it is a "macroeconomic concern". This is why, key C, is correct.

A) Monopolies result in market failure where a single seller or producer dominates the market, leading to reduced competition and potentially higher prices and reduced consumer welfare. State intervention through antitrust laws and regulation is often justified to address this market failure and promote fair competition.

B) Pollution of rivers is an example of a negative externality, where the actions of producers or consumers result in costs to others not directly involved in the market transaction. Pollution is a market failure because polluting firms may not bear the full costs of their actions, leading to an inefficiently high level of pollution. State intervention through environmental regulations, taxes, or cap-and-trade systems (pollution permits) is justified to internalize the external costs and address this market failure.

D) Public goods are a type of market failure because they are non-excludable and non-rivalrous, meaning individuals cannot be excluded from using them, and one person's use does not diminish their availability to others. Private markets may therefore not provide public goods due to the free-rider problem. State intervention is justified to provide public goods and ensure they are available for everyone in the society.

15-D:

D- Emphasis should be placed on the word "may" here. Creation of an equitable situation may increase efficiency because when individuals feel they are being treated fairly, they are more likely to be motivated and encouraged in their work, leading to increased productivity.

It might even not in some cases, because mainly richer people are seen to contribute more to economic efficiency. But since this option uses "may", this will be correct

Why not C and B? It is not necessary that for "equity" to be achieved, incomes should be re-distributed "equally". This is because equity and equality are two different concepts. Equity believes in a just and fair approach considering the unique needs of each individual, whereas equality is based on equal opportunities for everyone.

B - Positive economic judgements are based on facts. This is incorrect because efficiency and equity decisions can also be based on normative economic judgements or opinions. This is because equity decisions can be influenced by subjective evaluations based on personal values, and cultural and social norms too. The definition of what is fair, just and right may vary from person to person because of this.

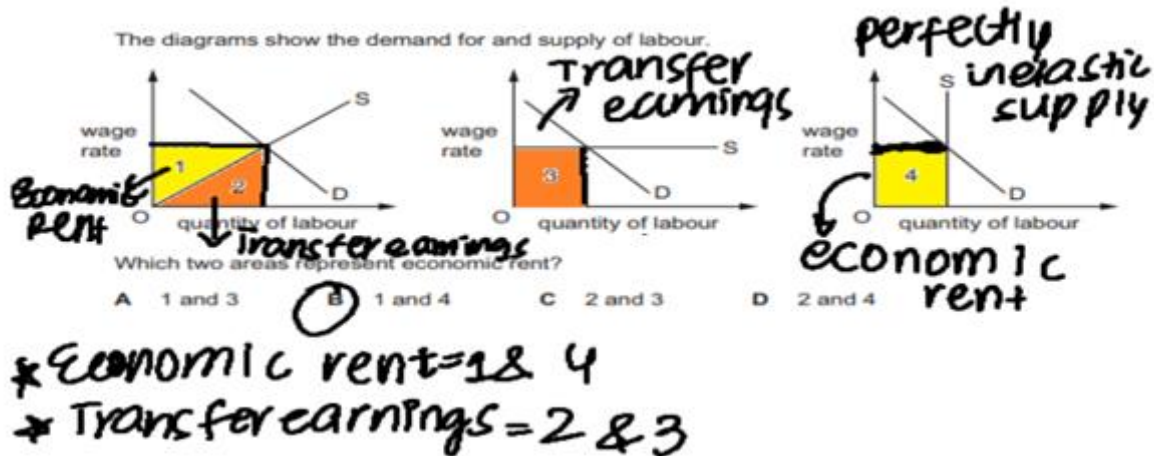
16-B:

Transfer earnings are the minimum income a worker needs in order to supply their labor. Economic rent is the extra income a worker receives – above the minimum level they need in order to work.

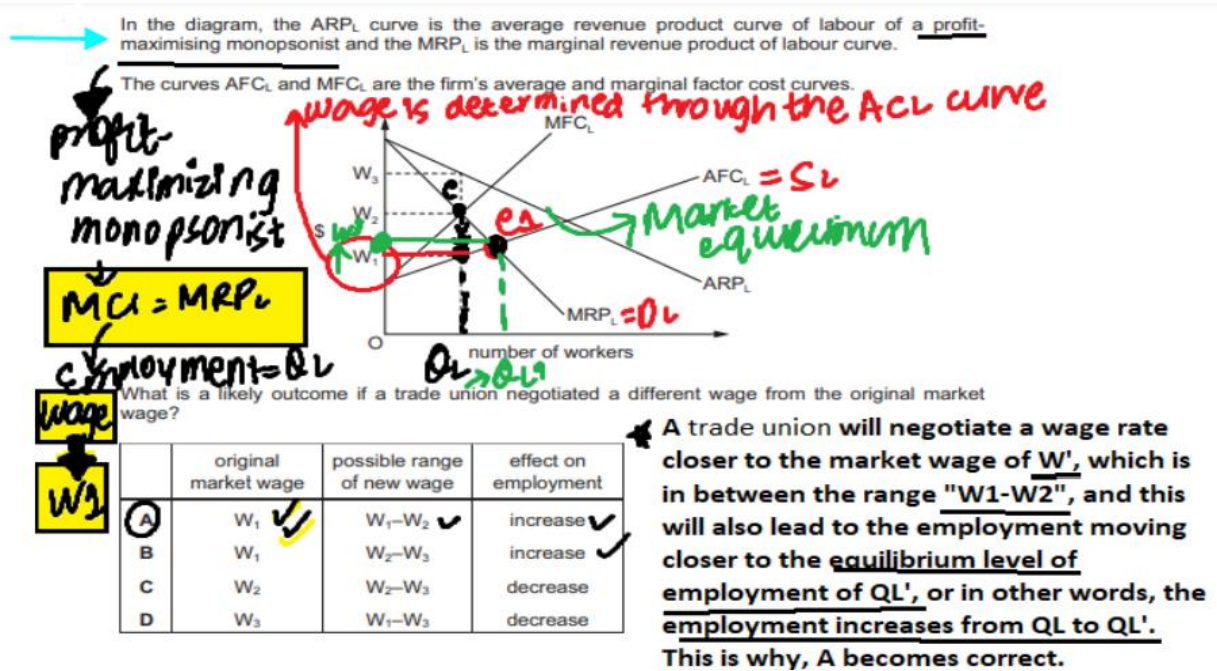
→ The area below the supply curve of labor represents “transfer earnings”.

→ The area above the supply curve of labor and below the wage rate represents the “economic rent”

The diagram below explains why, key B, will be correct.



17-A:



18-C:

If wages are set at a national level, it means that wages are the same across all regions, including depressed and prosperous ones. This would reduce the incentive for workers to move from a depressed region to a prosperous one, as they would not experience a significant increase in wages. **This explains why key C is correct.**

A) A developed market in rented accommodation in the prosperous region would actually increase labor mobility, as it would make it easier for workers to find housing and relocate.

B) A high rate of job turnover in the prosperous region would also increase labor mobility, as it would create more job opportunities and encourage workers to move to the region.

D) The withdrawal of a statutory minimum wage would likely lead to lower wages in the depressed region, making it even more attractive for workers to move to the prosperous region.

19-A:

Labor Productivity = Output / Employment

Now, let's analyze the scenario where labor productivity increases more than real national output:

1. Output increases: This means the numerator (O) in the labor productivity formula is increasing.
2. **Employment decreases: This means the denominator (E) in the labor productivity formula is decreasing.**

When the numerator (output) increases and the denominator (employment) decreases, the labor productivity ratio (O/E) will increase.

B) A deficit in the balance of trade would not directly explain why labor productivity increases more than an increase in real national output.

C) An increase in real wages would not explain why labor productivity increases more than an increase in real national output.

D) An increase in the rate of inflation would not explain why labor productivity increases more than an increase in real national output.

20-B:

The table shows the annual percentage (%) change in China's GDP and the contribution of some of its component parts between 2006 and 2012.

	2006	2007	2008	2009	2010	2011	2012
GDP	12.7	14.2	9.6	9.2	10.4	9.3	7.7
consumption	5.1	5.6	4.2	4.6	4.5	5.2	4.2
investment	5.5	6.0	4.5	8.1	5.5	4.5	3.9
net exports	2.1	2.6	0.9	-3.5	0.4	-0.4	-0.4

A – We are only given information about net exports, based on which, we cannot conclude the information of current account.

B – It can be seen that China's economic growth rate measured through changes in GDP fell in four years, more than it rose, since it rose in two years.

C – The overall growth rate is greater overall in capital goods than in consumer goods since the overall percentage increase in investment is greater than the overall percentage increase in consumption.

Overall % increase in consumption: $5.1\% + 5.6\% + 4.2\% + 4.6\% + 4.5\% + 5.2\% + 4.2\% \Rightarrow 33.4\%$

Overall % increase in investment: $5.5 + 6.0 + 4.5 + 8.1 + 5.5 + 4.5 + 3.9 \rightarrow 38\%$

D – Based on information of GDP and changes in components of GDP, we cannot conclude if standard of living rose at a decreasing rate.

21-C:

A) A decreased population is not a common factor in rapid development. In fact, many rapidly developing countries have experienced population growth due to immigration. A larger population can provide a workforce, drive consumer demand, and foster innovation.

B) Increased emphasis on government planning is not a common factor in rapid development. **Many successful countries have adopted market-oriented policies and reduced government intervention.**

C) Increased openness to international trade and investment is a common factor in rapid development. Countries that adopt policies promoting economic openness tend to experience faster growth, improved productivity, and increased job creation.

D) Increased unemployment is not a characteristic of rapid development. In fact, job creation and reduced unemployment are often outcomes of successful development strategies. Rapidly developing countries typically experience increased labor demand, driven by growing industries.

22-C:

We are using real GDP per capita data to compare living standards between two countries. Since the data is already adjusted for population size (per capita) and inflation (real data), it won't exaggerate the differences in living standards, thus making key A and B incorrect.

Now, option D states that the proportion of country Y's industry owned by foreign firms is higher. This means that most of the profits in country Y are repatriated abroad. Therefore, country Y's average living standard should be lower.

If we adjust for this data, the gap between country Y and country X will increase, as country Y's living standard will be lower. So, this represents that the statistics didn't exaggerate the gap; instead, the statistics understated the gap, which is actually larger.

Option C states that the proportion of services that people provide for themselves is higher in country Y. If there are such unreported activities in country Y, then country Y's GDP will be understated. When I adjust for this, country Y's living standard should be higher, and the gap between X and Y should be smaller.

However, the statistics showed a larger gap, which means that the difference in living standards is exaggerated.

This explains why key C will be correct.

23-D:

- Increasing income tax (A) could lead to a reduction in disposable income, because of which, consumer spending will decrease. This will result in a decrease in Aggregate demand, which will hinder economic growth.
- A higher birth rate (B) might increase the population size, which might reduce GDP per head, which denotes economic growth.
- A budget surplus (C) might indicate reduced government spending since it represents that tax revenue exceeds government spending which means that Aggregate demand is falling.
- An increase in training can increase productive capacity or Long-run Aggregate Supply by enhancing the skills and knowledge of workers, allowing them to produce more and better quality goods and services. As workers become more proficient, they can operate more efficiently, reduce waste, and improve productivity. **This explains that it can help economic growth in developing countries in the long-run, i.e. over the next five years.**

24-D:

The theory of the multiplier explains how changes in **autonomous injections**, for **example**: investment can lead to changes in national income. The multiplier effect suggests that an increase in investment leads to an initial increase in income, which then generates additional rounds of spending, leading to further increases in income. **This process continues, with each round of spending generating additional income, until the total increase in income is a multiple of the initial increase in investment.**

Here's how it works:

1. Initial increase in investment → Increase in income
2. Increase in income → Increase in consumption
3. Increase in consumption → one person's spending becomes the other person's income → Additional rounds of spending
4. Additional rounds of spending → Further increases in income

employment level of income

$\uparrow AE: U$
 $\uparrow Y: V$
 $+G (AE) \downarrow$

deflationary gap $\Rightarrow VU$

$\rightarrow PQ \rightarrow -ve$ output gap

In a closed economy

$AE = C + I + G$

Note: The deflationary gap is calculated by seeing how much does Aggregate expenditure fall short of real national income at potential level of output i.e. Q.

27-A:

A balance of payments surplus means that a country is receiving more foreign currency than it is spending, which leads to an increase in the country's foreign exchange reserves. When this is converted into domestic currency, money supply increases.

B. A government budget surplus would actually reduce the money supply, as the government would be reducing its spending.

C. If commercial bank depositors withdraw their cash, the money supply in the economy would decrease. When depositors withdraw their cash, the deposits in commercial banks decrease. With reduced deposits, banks have less money to lend to borrowers. With less lending, less new money is created. As a result, the overall money supply in the economy decreases.

D. The recall of loans by commercial banks would reduce the money supply, as the banks would be reducing their lending and increasing their reserves.

28-D:

When unemployment increases, the government's revenue from taxes decreases, and its expenditure on unemployment benefits and other social programs increases. **This leads to an increase in the government budget deficit. This explains why key D will be correct.**

A. A switch from direct to indirect taxation is not a direct consequence of increasing unemployment.

B. Demand-pull inflation is unlikely to occur when unemployment is rising, as higher unemployment typically leads to lower consumption as people have less incomes, which consequently results in lower aggregate demand and reduced inflationary pressures.

C. An increase in the balance of trade deficit is not a consequence of rising unemployment. This is because, when individuals have less incomes, their consumption of imported goods will fall. This will translate into a fall in import expenditures, thereby translating into a reduction in BOT deficit.

29-D:

A decrease in interest rates will result in a fall in savings, as the return on saving reduces. This will encourage individuals to consume more.

A decrease in interest rates makes borrowing cheaper, which can lead to an increase in consumption as individuals are more likely to take out loans to finance their consumption of big ticket items.

This explains why consumption will increase, thereby making key D will increase.

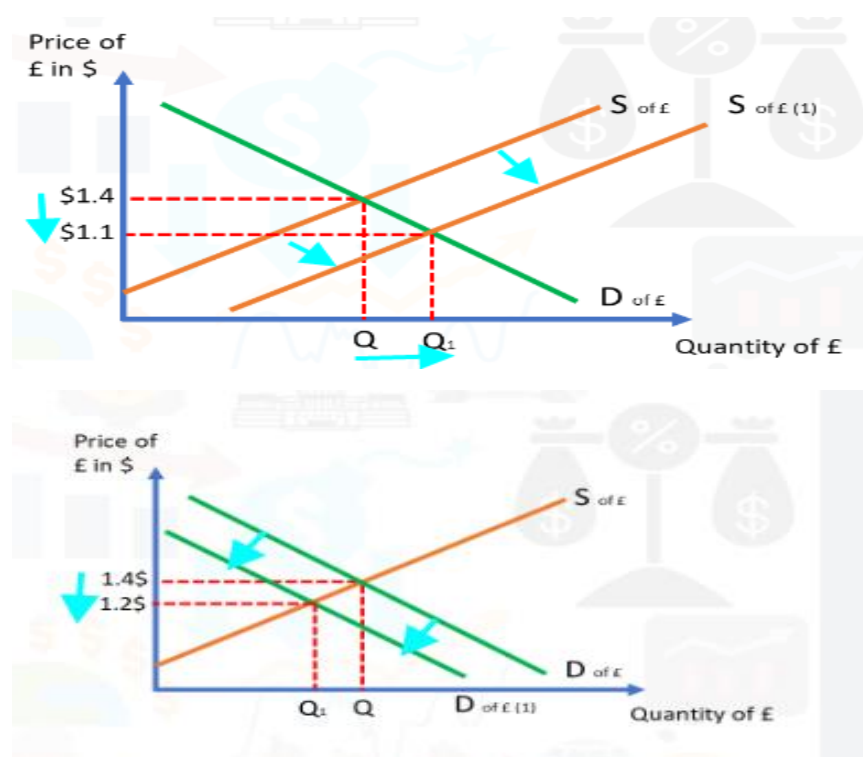
A. A decrease in interest rates is likely to reduce costs of borrowing which will result in an increase in investment, not a decrease.

B. A net capital inflow is unlikely to result from a decrease in interest rates, as lower interest rates can make domestic investment less attractive to foreign investors.

C : A lower interest rate can lead to a depreciation of the currency because there is generally a positive relationship between interest rate and exchange rate.

- ➔ With lower interest rates, domestic investment becomes less attractive to foreign investors and these hot money outflows lead to a decreased demand for the domestic currency.
- ➔ Lower interest rates reduce the incentive for individuals and businesses to invest in the domestic economy and since they are likely to sell their currency to buy foreign currency, this will lead to an increase in the supply of domestic currency in the foreign exchange market.

This is how, the currency depreciates.



30-A:

An increase in interest rates will make bonds more attractive to financial investors, leading to a decrease in share prices. This is because:

- Higher interest rates increase the return on bonds, making them more attractive to investors.
- As a result, investors may shift their investments from shares to bonds, leading to a decrease in demand for shares and a subsequent decrease in share prices.

Here's why the other options are incorrect:

B. An increase in interest rates is likely to reduce inflationary pressure, not increase it.

C. An increase in interest rates can lead to higher borrowing costs for companies, which can reduce company profits, not increase them.

D. An increase in interest rates will make saving more attractive, not less attractive. Higher interest rates increase the return on savings, making them more appealing to investors.

This explains why key A will be correct.